CHAPTER 1

INTRODUCTION

A QR code is much like a bar code, except that because it carries data in two dimensions (horizontally and vertically), it is able to hold much more information than a one-dimensional bar code. In fact, compared to a bar code's 20 alphanumeric character limit, a QR code can hold thousands of characters of data. As a result, a QR code can be used to share multimedia content, a landing page or an entire e-book. However, QR codes can do much more than that – they can actually direct a phone to perform certain actions. For example, a theatre company might provide a QR code that not only sends the person who scans it to the company's website for show times and ticket information, but also embeds information about the dates, times and locations of upcoming shows into the phone's calendar. QR codes were designed by DENSO WAVE in Japan and first came into use in 1994. Although the term "QR code" is a registered trademark, the technology itself has not been patented and is therefore available for anyone to use. One key benefit of these codes for the Japanese is that they are one of the most efficient ways of encoding Japanese characters. However, their high data capacity, small print-out size and resistance to dirt and damage (QR codes can still be read when up to 30 percent of the code has been corrupted) ensured the proliferation of this technology to other applications. In addition, unlike bar codes, which can only be read by a special scanner, many smartphones are able to scan QR codes, making them an excellent way to provide large amounts of information in a very small space. A QR code consists of black squares arranged in a square grid on a white background, including some fiducial markers, which can be read by an imaging device such as a camera, and processed using Reed–Solomon error correction until the image can be appropriately interpreted. The required data is then extracted from patterns that are present in both horizontal and vertical components of the image.

CHAPTER 02

LITERATURE SURVEY

Quick Response code is usually authenticated with the help of the camera of one's mobile phone. QR codes can easily scanned through mediums like Tablets, laptops and personal computer desktops. The system automatically generates the ID of the user and its password. The characteristic which makes QR codes stand out is they can still be scanned even if they are partially damaged. QR codes are a 2 dimensional printing code on a paper or a screen which makes it pretty vulnerable from various type of cyber-attacks. It can harm your device by unknowingly directing you to a virus contaminated page or website. To avoid this, one must verify the origin of a particular QR code and must have a full understanding of the data type of that particular QR code. There are many attacks involving QR codes as well as their solutions. QR codes are becoming quite popular nowadays because of the rapid increase in smart devices by the normal people around the world. Obviously, 2D QR code is way better and store huge amount of encoded information compared to the old traditional 1D codes. People are using smartphones to do authentications and for this the QR codes are the most ideal way to do it. Many types of QR codes are getting popular nowadays including logo QR code, encrypted QR code, iQR Code etc. QR codes are becoming popular day by day in the upcoming generation as it offers way easier authentication that the traditional old fashioned user id and password. QR codes offers many advantages such as greater storage capacity, fast readability, 360 degree reading, small print size, error correction, support for more languages and durability against soil and damage. Many firms who are relatively new in the online business are tend to use these codes instead of normal login process. To fix the QR information / security issue, Xiaohe Cao proposed a safe QR code scheme based on visual cryptography. The security problem of QR code is severe, such as data loss and data tampering as the implementation of QR Code is wide enough. The QR code is spilt into two shared pictures which will be transmitted singly. The development of the two shared pictures is based on the pseudo-random matrix, i.e. the pixels are determined by the pseudo-random matrix values in the two shared pictures. The two images

shared can only be stacked to revive the information. Simulation output demonstrates that the picture of the QR code can be masked well and can be efficiently reconditioned. Peter Kieseberg has examined how both automated systems and human interaction can be attacked using QR Codes. As the encoded data is meant to be machine readable only, one cannot differentiate between a legitimate and a harmful corrupted QR code. While automated readers are verymuch endangered to SQL injections and command injections, individuals might be prone to phishing attacks. Peter Kieseberg contribution is a survey of the QR code as an attack vector, demonstrating different attack plans for the attackers to read and explore the implications

CHAPTER 03

PROBLEM STATEMENT

Generating a QR code and developing a QR code reader which will be used for business process integration. We are going to develop an android application which would be able to read the QR code. This project is intended to small businesses which would use this technology for marketing purpose

CHAPTER 04

EXISTING SYSTEM

1. QR Code Generators Online: There are numerous websites that offer free QR code generation services. You can simply input your data (such as a URL, text, or contact information), and the website will generate a QR code for you to download. Examples include qr-code-generator.com and qrstuff.com.

2. QR Code Generator Software: There are downloadable software applications for Windows, macOS, and Linux that allow you to create QR codes offline. Examples include QR Code Generator for Mac and QR Code Studio for Windows.

3. QR Code Generator Libraries: If you're a developer, you can use programming libraries to generate QR codes within your applications. Popular libraries include ZXing (for Java), qrcode.js (for JavaScript), and qrencode (for C/C++).

4. Mobile Apps: There are many mobile apps available for both Android and iOS that can generate QR codes. Some popular choices include QR Droid (Android) and QR Reader (iOS).

5. Integrated QR Code Generators: Some applications and platforms, like Microsoft Word and Google Sheets, have built-in QR code generators. You can insert a QR code directly into your documents or spreadsheets.

CHAPTER 05

PROPOSED SYSTEM

The development of QR based new system will allow technology savvy users to just scan QR codes to complete end to end business process flow execution to complete task at hand. This kind of system is cost effective at the same time available 24 / 7 to do a business transaction. Small business owners know that they need all the marketing that they can get. It brings in sales, allow people to know about them and even introduce these businesses to new customers. A relatively exciting field right now is mobile marketing, which is both very affordable and very effective right now. Within a few years, QR codes will be an integral part of retail spaces and essential for experiential marketing campaigns. Now QR Code is seen and used everyday everywhere in Japan for the following reasons:

1. Several characteristics superior to linear bar codes: much higher data density, support Kanji/Chinese character, etc.

2. It can be used by anybody free of charge as Denso has released the patent into the public domain.

3. Data structure standard is not prerequisite for current usages.

4. Most mobile phones in Japan equipped with cameras that enable reading of QR Codes can access Internet addresses automatically by simply reading a URL encoded in the QR Code.

CHAPTER 06

SYSTEM REQUIREMENT SPECIFICATION

6.1 Functional specification

1.Mobile Module: The mobile application runs as a background process in the customer's mobile phone and it will scan the QR code and will fetch the stored information. There are four sub modules

2. Electronic Items: In this module the mobile application will read the QR code and will fetch the information stored in the QR code .As now for this module we are going to store information about the electronic items that is it's specifications , cost in which shops is that item available, address of the shop etc.

3. Hotel Information: In this module will read the QR code and will fetch the information stored in the QR code. This module will store the hotel information that is booking available in the hotel, cost of the room etc.

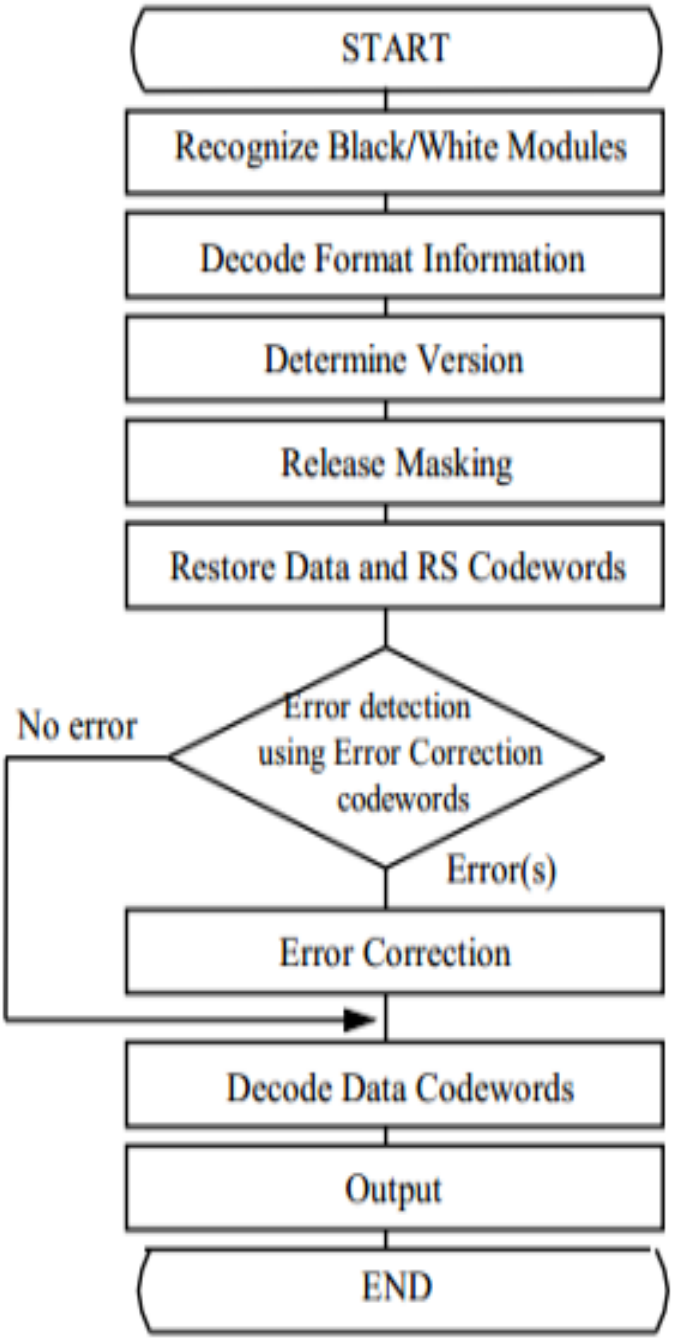
4. Famous Places: In this module will read the QR code and will fetch the information stored in the QR code. It will include the famous places, its map (how to reach that place), mode of transport etc.

5. Bus Time Table: In this module will read the QR code and will fetch the information stored in the QR code. It will include the source, destination, the cost of ticket and the bus time table.

CHAPTER 07

DESIGN

FLOW CHART OF QR CODE GENRATOR



CHAPTER 08

CODING FOR QR CODE GENRATOR

8.1 CODING

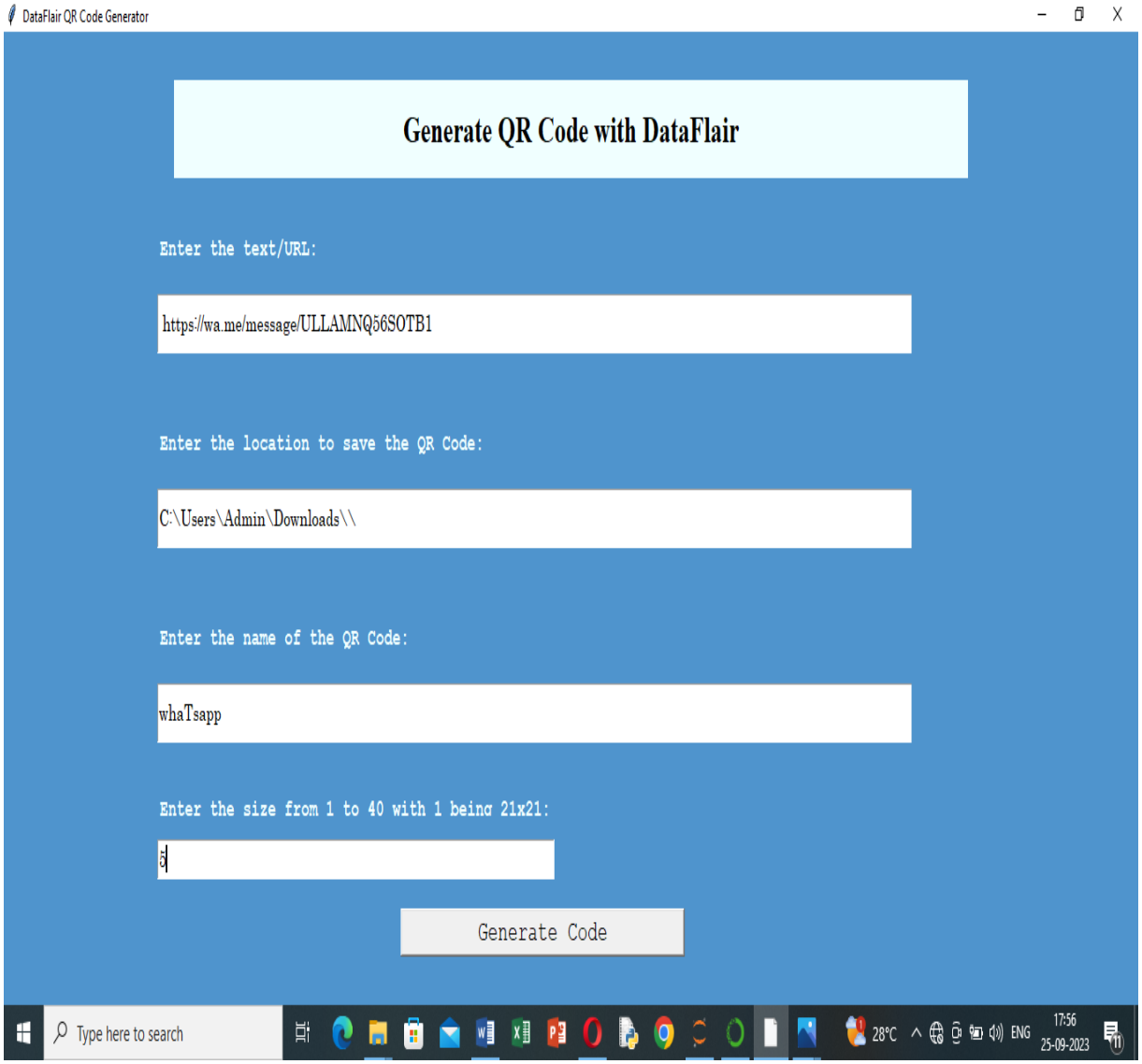
```
import qrcode
from tkinter import *
from tkinter import messagebox
#Creating the window
wn = Tk()
wn.title('DataFlair QR Code Generator')
wn.geometry('700x700')
wn.config(bg='SteelBlue3')

#Function to generate the QR code and save it
def generateCode():
    #Creating a QRCode object of the size specified by the user
    qr = qrcode.QRCode(version = size.get(),
                box_size = 10,
                 border = 5)
    qr.add_data(text.get()) #Adding the data to be encoded to the QRCode object
```

```
  qr.make(fit = True) #Making the entire QR Code space utilized
  img = qr.make_image() #Generating the QR Code
  fileDirec=loc.get()+'\\'+name.get() #Getting the directory where the file has to be save
  img.save(f'{fileDirec}.png') #Saving the QR Code
 #Showing the pop up message on saving the file
 messagebox.showinfo("DataFlair QR Code Generator","QR Code is saved successfully!")

#Label for the window
headingFrame = Frame(wn,bg="azure",bd=5)
headingFrame.place(relx=0.15,rely=0.05,relwidth=0.7,relheight=0.1)
headingLabel = Label(headingFrame, text="Generate QR Code with DataFlair", bg='azure',
font=('Times',20,'bold'))
headingLabel.place(relx=0,rely=0, relwidth=1, relheight=1)

#Taking the input of the text or URL to get QR code
Frame1 = Frame(wn,bg="SteelBlue3")
Frame1.place(relx=0.1,rely=0.15,relwidth=0.7,relheight=0.3)
   lable1 = Label(Frame1,text="Enter the text/URL:
",bg="SteelBlue3",fg='azure',font=('Courier',13,'bold'))
```

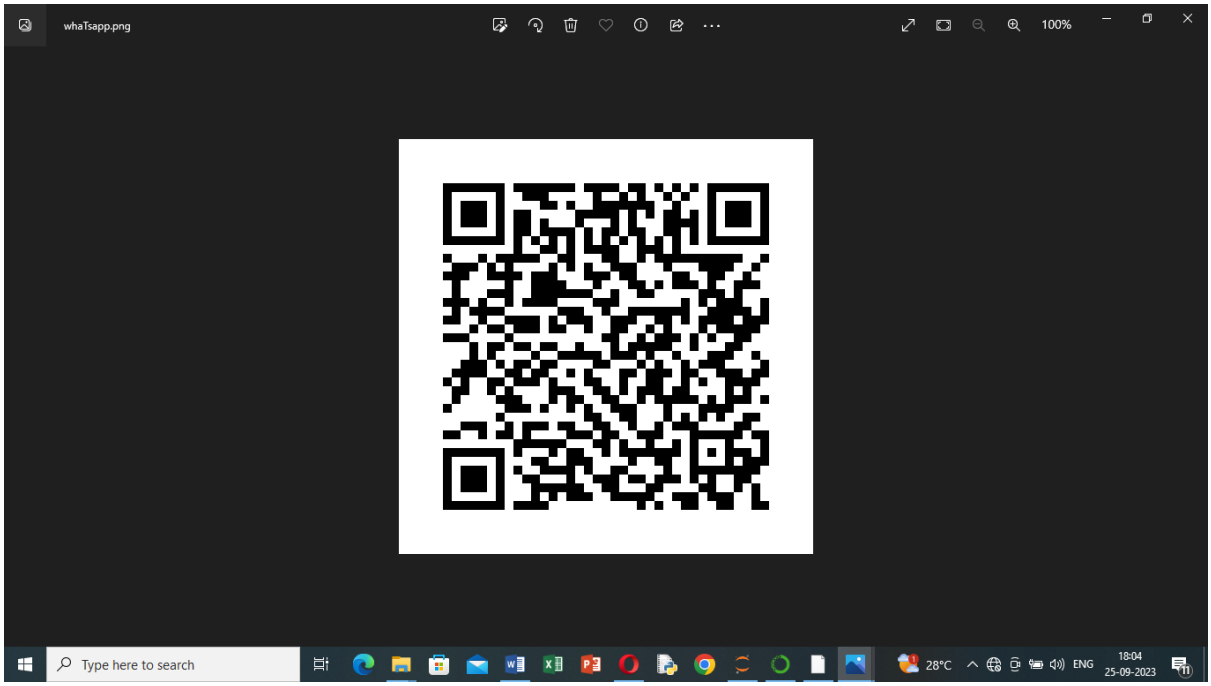
```
lable1.place(relx=0.05,rely=0.2, relheight=0.08
text = Entry(Frame1,font=('Century 12'))
text.place(relx=0.05,rely=0.4, relwidth=1, relheight=0.2)

#Getting input of the location to save QR Code
Frame2 = Frame(wn,bg="SteelBlue3")
Frame2.place(relx=0.1,rely=0.35,relwidth=0.7,relheight=0.3)
lable2 = Label(Frame2,text="Enter the location to save the QR Code:
",bg="SteelBlue3",fg='azure',font=('Courier',13,'bold'))
lable2.place(relx=0.05,rely=0.2, relheight=0.08)

loc = Entry(Frame2,font=('Century 12'))
loc.place(relx=0.05,rely=0.4, relwidth=1, relheight=0.2)

#Getting input of the QR Code image name
Frame3 = Frame(wn,bg="SteelBlue3")
Frame3.place(relx=0.1,rely=0.55,relwidth=0.7,relheight=0.3)

lable3 = Label(Frame3,text="Enter the name of the QR Code:
",bg="SteelBlue3",fg='azure',font=('Courier',13,'bold'))
lable3.place(relx=0.05,rely=0.2, relheight=0.08)
```

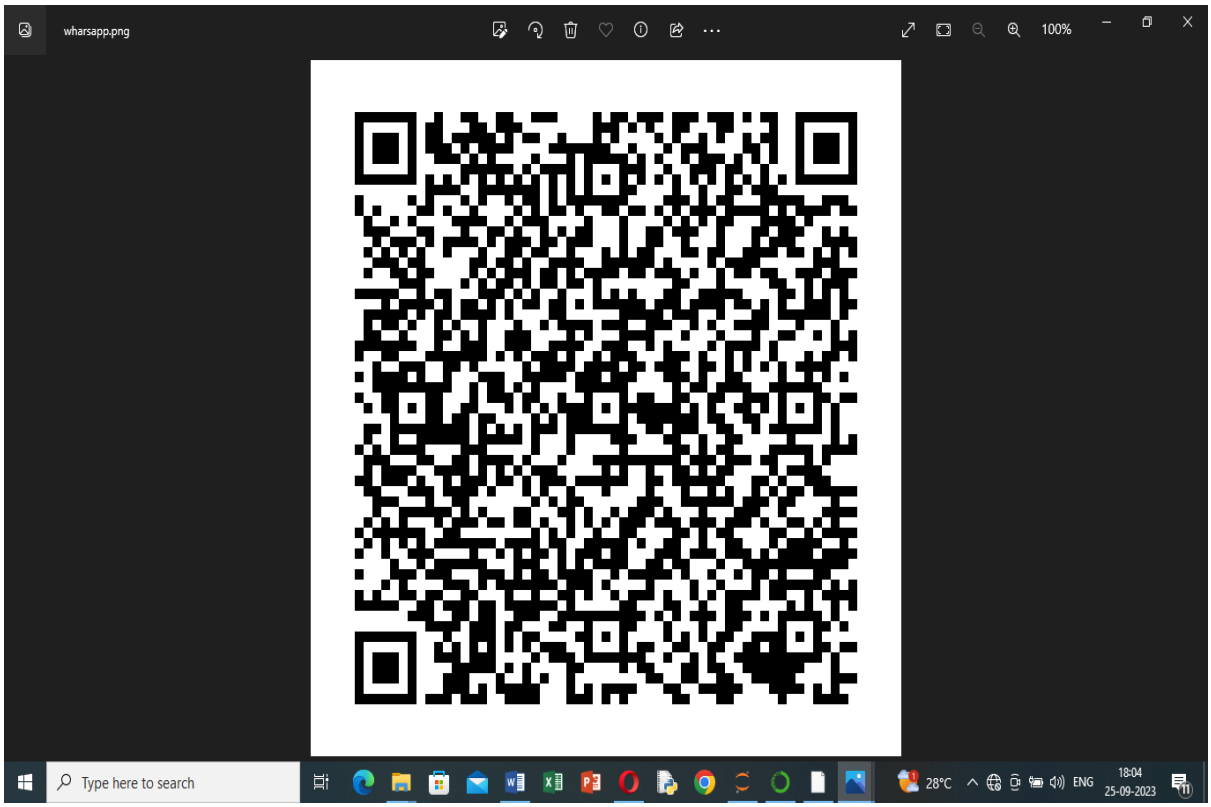
```
name = Entry(Frame3,font=('Century 12'))

name.place(relx=0.05,rely=0.4, relwidth=1, relheight=0.2)


#Getting the input of the size of the QR Code

Frame4 = Frame(wn,bg="SteelBlue3")

Frame4.place(relx=0.1,rely=0.75,relwidth=0.7,relheight=0.2)


lable4 = Label(Frame4,text="Enter the size from 1 to 40 with 1 being 21x21:
",bg="SteelBlue3",fg='azure',font=('Courier',13,'bold'))

lable4.place(relx=0.05,rely=0.2, relheight=0.08)


size = Entry(Frame4,font=('Century 12'))

size.place(relx=0.05,rely=0.4, relwidth=0.5, relheight=0.2)


#Button to generate and save the QR Code

button = Button(wn, text='Generate
Code',font=('Courier',15,'normal'),command=generateCode)

button.place(relx=0.35,rely=0.9, relwidth=0.25, relheight=0.05)


#Runs the window till it is closed manually

wn.mainloop()
```

8.2OUT PUT


DataFlair QR Code Generator
Generate QR Code with DataFlair
Enter the text/URL:
https://wa.me/message/ULLAMNQ56SOTB1
Enter the location to save the QR Code:
C:\Users\Admin\Downloads\\
Enter the name of the QR Code:
whaTsapp
Enter the size from 1 to 40 with 1 being 21x21:
5
Generate Code
Type here to search
28°C
ENG
17:56
25-09-2023

SCREENSHORTS





CHAPTER 09

9.1 SOFTWARE REQUIREMENTS

Operating system	Windows 10 and above
Programing language	Python
Laibraries	Qrcode, tkinter

CHAPTER 10

CONCLUTION

This project will benefit small & medium business to adopt new technology and increase the consumer traffic. It is cost effective solution to medium sized business as compared to the individual hosted solution. Based on this research of this thesis, the QR Code can be regarded as a considerable value for organizations. With the development of QR code, QR Code is widespreadly used in a number of fields, for instance in newspapers, magazines, public advertisements, business advertisements, food, drinks, restaurants, clothes stores, social network applications, enterprises, and government. Nevertheless, QR Code is relatively new for most companies. While QR Code is a competitive marketing tool today, it can also be regarded to provide commercial opportunities in different business areas in the future. As a result, the concept of QR Code was needed to be understood for individuals and organizations. To summarize the findings of the research, QR Code is a two-dimensional barcode that can be scanned from vertical as well as horizontal angles. QR Code was created in 1994 by Wave Denso. QR Code is free of charge and there was no potential problem existed for both organizations and individuals. QR Code was used widely with the development of smart-phones. QR Code is divided into six types which refer to QR Code Model one, Model two, Micro QR Code, IQR Code, SQRC and LogoQ. Moreover, QR Code has a number of characteristics, which include high capacity encoding of data, kanji and kana capability, small printout size, error and correction capacity, readable form any direction in 360 degree and structure appending feature. The technologies of QR Code were introduced in this thesis as well. QR Code contains forty versions of QR Code, and the structure of QR Code, which is divided into two segments. There are function patterns, which are composed of the finder pattern, separator, timing patterns and alignment pattern. Moreover, the function patterns focus on the positioning. Another part is encoding region, which involves format information, version information, data and error correction code-words. Besides, encoding and decoding procedure were also displayed in this thesis. Guidebooks of generating QR Code and reading QR Code were listed in this thesis for guiding people to QR Code implementation as well.

CHAPTER 11

FUTURE SCOPE

By using QR code reader we will read the QR Code extract the id and fetch the details about that entity from the database and present it to the user. We are also providing search facility so that the user is not completely depended on the QR code and can search the entity directly.

1. Enhanced Security: QR codes could incorporate more advanced encryption and security features to protect sensitive information, making them suitable for secure transactions and data sharing.

2. Augmented Reality (AR): Integration with AR technology could allow for interactive QR codes, where scanning a code can trigger immersive experiences, such as product demos or virtual tours.

3. Dynamic QR Codes: These codes can be updated with new information, enabling real-time tracking and management. They are valuable for marketing campaigns, inventory control, and event management.

4. Personalization: QR codes might be customized with unique designs and branding elements, making them more visually appealing and recognizable.

5. IoT Integration: QR codes could play a significant role in connecting physical objects to the Internet of Things (IoT), facilitating device setup and data sharing.

CHAPTER 12

REFERENCE

1. Alexandre Alapetite, “Dynamic 2D-barcodes for multi-device Web session migration including mobile phones”,ACM Digital Library, Springer-Verlag London, UK , Volume 14 Issue 1, January 2010 .

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3. Judy C. R. Tseng, Sunny Y. Y. Hsu, Gwo-Jen Hwang,” A collaborative ubiquitous learning platform for computer science education”, ITIiCES, 2009